STOCHASTIC METHODS IN WATER RESOURCES

Unit 4: Time series analysis
Lecture 15: Special properties and models

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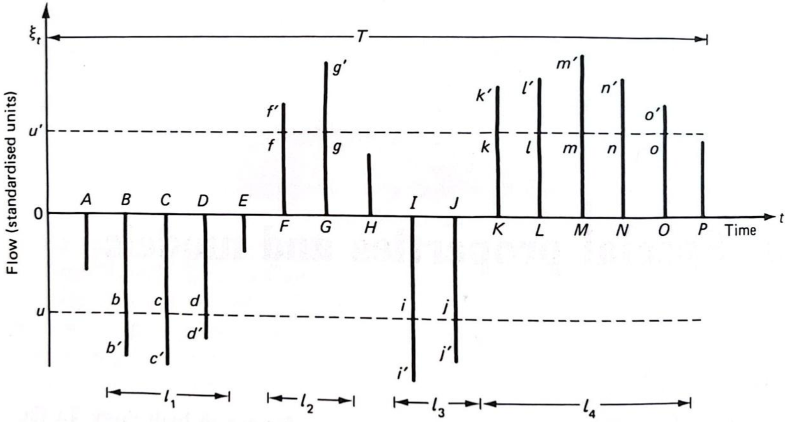
Runs and crossing properties

Generalities

- There does not appear to be universally accepted method of defining **droughts** in hydrometeorology. However, one criterion which has practical significance pertains to the deficits with respect to (or excursions below) a specified level or rate of flow in a river.
- The duration of a deficit is known as a **run length**. In general, a **run** is defined as a sequence of observations of the same kind preceded and succeeded by one or more observations of a different kind.
- Corresponding to the run length, the magnitude of the deficit is known as a run sum. Such characteristics of a time series belong to a general class called crossing properties.
- To explain these specifically, consider the crossing level u in the figure bellow which shows a section $(0, T)$ of a stationary discrete time series ξ_t with zero mean and variance σ_ξ^2.

Runs and crossing properties

Generalities



Crossing properties of a discrete time series, deficit run lengths l_1, l_3 at level u; surplus run lengths l_2, l_4 at level u'; downcrossings of level u at B and I with deficit run sums $bb' + cc' + dd'$ and $ii' + jj'$ respectively; upcrossings of level u' at F and K with surplus run sums $f'f + g'g$ and $k'k + l'l + m'm + n'n + o'o$ respectively.

Runs and crossing properties

Generalities

- Downcrossings of this level occur a times B and I because the flows at these times are less than u, whereas flows at the antecedent times A and H respectively are greater than u.
- The corresponding deficit run lengths are l_1 and l_3. Also, the quantities $bb' + cc' + dd'$ and $ii' + jj'$ are known as the deficit run sums.
- Here, u can represent the level in a river which should be maintained in order to meet water supply demands. Alternatively, u may be relevant to the minimum rate of flow required for pollution control. Again it could be the lowest acceptable level for navigation purposes.
- Then the deficit run sums represent the quantities that need to be released, for example, from a regulating reservoir. Furthermore, the design and operation of hydroelectrici schemes and the sizing of pumps are related to crossings and run sums.
- Likewise, the durations and magnitudes of flows above a fixed high levels such as u' in the figure above, known as the surplus run lengths and the sums with respect to this level, are of importance for flood control purposes.
- In the case shown, upcrossings of level u' occur at times F and K, and the surplus run lengths are l_2 and l_4. The run sums, given by $f'f + g'g$, $k'k + l'l + m'm + n'n + o'o$ and so on, can be regarded as the storage reservation that needs to be made in a reservoir for the abatement of floods downstream.
- Number of level crossings too have an important physical relevance. The analysis of rare floods and, in general, the theory of extreme values are relevant to the crossings of levels remote from the mean.
- In addition, the mean number of crossings which characterise a time series are of interest to river engineers.

Runs and crossing properties

Theoretical aspects

The theory of crossings says that the expectations of the number of downcrossing of a level u (which is equal to the expectation of the number of upcrossings of the same level) of a stationary normal stochastic process ξ_t with zero mean and variance σ_ξ^2 in an interval $0 \leq t \leq T$ is given by

$$E[N_T^-(u)] = T(2\pi)^{-1} \left(\frac{\lambda_2}{\sigma_\xi^2} \right)^{1/2} e^{-\frac{u^2}{2\sigma_\xi^2}} \quad (1)$$

in which λ_2 is the second moment of the spectral density function $f(\omega)$, where ω is the frequency in cycles per unit time; in general,

$$\lambda_r = \int_0^\infty \omega^r f(\omega) d\omega$$

Also, if $C''(0)$ is the second derivative of the covariance function at the origin

$$\lambda_2 = -C''(0) \quad (2)$$

It follows that, for the mean level ($u = 0$), $E[N_T^-(0)] = T(2\pi)^{-1} \left[-C''(0)/\sigma_\xi^2 \right]^{1/2}$ in this way, the mean number of crossings of the zero level characterises a particular time series.

The applicability of equations such as equation 1 is unfortunately limited because of the fundamental non-normal nature of hydrologic time series. Another apparent disadvantage is that practical investigations are made in discrete time.

Runs and crossing properties

Theoretical aspects

If we consider again the discrete case shown in the figure above, let $L^{-}(u)$ and $S^{-}(u)$ denote a deficit run length and deficit run sum, respectively, in the interval $(0, T)$, where T is large. As before, $N_T^{-}(u)$ denotes the number of downcrossings in $(0, T)$; this is also the number of deficit runs in the same interval. If $Pr[\xi_{t-1} > u, \xi_t < u]$ is the probability of a downcrossing at time t. The expected number of downcrossings (or upcrossings in $(0, T)$) is

$$E[N_T^{-}(u)] = Pr[\xi_{t-1} > u, \xi_t < u]T \tag{3}$$

For a discrete time series we can give the expected values of the deficit run length and the deficit run sum as follows.

$$E[L^{-}(u)] = \frac{Pr[\xi_t < u]}{Pr[\xi_{t-1} > u, \xi_t < u]} \tag{4}$$

and

$$E[S^{-}(u)] = E[L^{-}(u)]E[u - \xi_t | \xi_t < u] \tag{5}$$

where $|$ means conditional to the inequality on the right.
One purpose this can serve is for comparing crossing properties in generated and historical data as a means of testing the validity of postulate models; the usefulness of such a procedure derives from the versatility of crossing properties.

Runs and crossing properties

Some properties of independent discrete series

Consider an independent and identically distributed discrete series X_t, $t = 1, 2, 3, \cdots$, which represents, say, the total precipitation at a particular site during the tth year. Let $F(x)$ denote the common distribution function of the X_t series. Also, let $q = F(u) = Pr[X \leq u]$, where u is a crossing level and $p = 1 - q$. Then, if $+1$ and -1 signify that X_t is greater than (or equal to) u and less than u respectively, the probability distribution of surplus run lengths is given by

$$\begin{aligned} Pr[L^+ = l] &= Pr\left[X_{i+2} = X_{i+3} = \cdots = X_{i+l} = 1, X_{i+l+1} = -1 | X_i = -1, X_{i+1} = 1\right] \\ &= \frac{p^l q^2}{pq} = p^{l-1} q \end{aligned} \tag{6}$$

for $l = 1, 2, 3, \cdots$ which is a geometric distribution. Also, it can be shown that the expected value and variance of surplus run lenghts is

$$E[L^+] = \frac{1}{q} \qquad var(L^+) = \frac{p}{q^2} \tag{7}$$

Note that for deficit run lenghts L^- the probabilities p and q should be interchanged in equations 6 and 7. A useful outcomes is that the mean run length is equal to 2 if the median value, for which $p = q = 0.5$, is taken as the crossing level.

Runs and crossing properties

Some properties of independent discrete series

An alternative way to estimate runs is to condition them on the number of X values which are represented by $+1$ or -1. The expected total number and variance of runs $N = N_i^+ + N_i^-$ are then given by

$$\begin{aligned} E[N] &= 1 + \frac{2nm}{n+m} \\ var(N) &= \frac{2nm(2nm-n-m)}{(n+m)^2(n+m-1)} \end{aligned} \tag{8}$$

where n and m represent the numbers of values above and below the chosen crossing level, respectively.

If long runs are observed in a given sequence, which is the same as having only few runs, it indicates some type of grouping due to a serial correlation, to a periodic movement or perhaps to a trend. In fact, a run test can be formulated to test for independence on the basis of equation 8. The significance test is made on the assumption that N is normally distributed, the null hypothesis being that the sequence is independent.

Range analysis

Hurst phenomenon

According to Hurst (1951), there is an approximate relationship between the rescaled range and the sample size. For an observed sequence $x_1, x_2, \cdots, x_N$ with $E[x] = \mu$, the range is defined by

$$r_N = max\left[0, max(x_1 + x_2 + \cdots + x_i - i\mu)\right] - min\left[0, min(x_1 + x_2 + \cdots + x_i - i\mu)\right] \tag{9}$$

This is sometimes referred to as the crude range.
It is advantageous for practical purposes to calculate the difference between the maximum d_N^+ and the minimun d_N^- of the accumulated departures from the sample mean $\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$. This gives the adjusted range r_N^*. Thus

$$r_N^* = max(x_1 + x_2 + \cdots + x_i - i\bar{x}) - min(x_1 + x_2 + \cdots + x_i - i\bar{x}) = d_N^+ + |d_N^-| \tag{10}$$

Note that, for $i = N$, the terms within each set of parentheses in equaution 10 sum to 0, but this does not necessarily apply to the corresponding terms in equation 9; hence, we have the addition of two zeros in the equation for r_N.

Range analysis

Hurst phenomenon

In order to compare the results from different observed sequences, the range is divided by the standard deviation S_N estimated by

$$s_N = \frac{1}{\sqrt{N}} \left[\sum_{i=1}^{N} (x_i - \bar{x})^2 \right]^{1/2}$$

to give the adjusted rescaled range

$$r_N^{**} = \frac{r_N^*}{s_N} \tag{11}$$

This division by the standard deviation is a unique feature of Hurst's work. Of greater significance is the relation

$$r_N^{**} = \left(\frac{N}{2} \right)^K \tag{12}$$

where K is deemed to be a constant for an observed sequence, which he established empirically. It was found that the exponent K varies from 0.46 to 0.96 with a mean of 0.73 and a standard deviation of 0.09 in the sequences which Hurst examined. For the individual groups of data analysed such as river flows we expect less variation.

Range analysis

Hurst phenomenon

Hurst also derived the theoretically expected value of the adjusted range. He used some coin-tossing experiments to show that, for independent standard normal variates, the asymptotic result

$$E[R_N^*] = \sqrt{\frac{\pi}{2}}\sqrt{N} \approx 1.253\sqrt{N} \tag{13}$$

holds.
Similarly, Feller (1951) gave an independent and rigorous derivations of the same result, which is applicable to stationary independent random variables with finite variance regardless of the distribution. Note that the result is valid only if N is large, say, greater than 1000. He also showed that the variance of R_N^{**} is

$$var(R_N^{**}) = \left(\frac{\pi^2}{6} - \frac{\pi}{2}\right) N = 0.074N \tag{14}$$

This holds approximately for $N \geq 50$.
Analogous to equation 12, the more general form

$$E[R_N^{**}] = cN^h \tag{15}$$

has been adopted by Mandelbrot and Wallis (1968) in which the exponent h is the Hurst coefficient and c is another constant. It was postulated that a natural process is characterized by a population value h; the small sample estimate of h is denoted by H. The fact that values of $H \neq 0.5$ are found in observed natural sequences whereas, for a normal independent or ARMA type process, the asymptotic value of h is 0.5 has become known as the **Hurst phenomenon**.

Estimation of Hurst coefficient h

- Mandelbrot and Wallis (1969) proposed a graphical procedure for estimating h, through a so-called pox diagram. This involves computations which use equation 11 of rescaled range $r^{**}_{t,n}$ based on different subsequence sizes $n \leq N$, from the same record, in contrast with the use of the complete sample size N as in the estimation of K from equation 12; t denotes the starting points of the subsequence which are also varied.
- Computed values are plotted on a double logarithm graph of $r^{**}_{t,n}$ againts n. If $\bar{R}^{**}_{n}$ denotes the estimated mean of the $r^{**}_{t,n}$ with n fixed and t variable, the estimated H of h is the slope of straight line fitted by eye to the plot of $\bar{R}^{**}_{n}$ against n for $n > n_0$; here $n_0 \approx 20$ is taken as the arbitrary upper limit of the so-called initial transient stage.
- For these computations, values of t and n are chosen so that the information abstracted is maximised as far as possible whilst minimizing the amount of overlapping between sequences.
- In general, estimates of h are biased because the slope h depends on n. That is, $E[\bar{R}^{**}_{n}] \approx n^{h(n)}$, whereas we are attempting to estimate $\lim_{n\to\infty}[h(n)] = h$, in which $h(n)$ is the local exponent.

Range and rescaled range in small samples

The interpretation of the Hurst phenomenon given above, which is the commonly accepted one, is undoubtedly important from theoretical considerations. However, for practical purposes knowledge of the small sample behaviour of the range statistics is required.

1. Independent identically distributed standard normal variate are considered. For this adjusted range, it has been proved that

$$E[R_N^*] = \sqrt{\frac{N}{2\pi}} \sum_{i=1}^{N-1} [i(N-i)]^{-1/2} \tag{16}$$

2. It has been stablished that, for the rescaled adjusted range

$$E[R_N^{**}] = \sqrt{\frac{1}{\pi}} \left[\frac{\Gamma((N-1)/2)}{\Gamma(N/2)}\right] \sum_{i=1}^{N-1} [(N-i)/i]^{1/2} \tag{17}$$

   Again, for a sequence of inputs from a gamma population with a two-parameter density function $f(x) = x^{\gamma-1}e^{-x/\lambda}/\lambda^{\gamma}\Gamma(\gamma)$ (where γ and λ are the shape and scale parameters respectively). Accordingly

$$E[R_N^*] = 2\gamma^{-1/2} N^{-\gamma N}\Gamma(\gamma N) \sum_{i=1}^{N-1} \left[\frac{i^{i\gamma-1}(N-i)^{\gamma(N-i)}}{\Gamma(i\gamma)\Gamma(N\gamma - i\gamma)}\right] \quad N = 1, 2, 3, \cdots \tag{18}$$

One important outcome of the theoretical work on the rescaled range as given by equation 17 is that for finite sequences of independent normal variates the exponent h is equation 15 is significantly greater than 0.5. The asymptotic value, $h = 0.5$, is reached only when N is much larger than the sample sizes found in practices.

Range analysis

Explanation of Hurst phenomenon and experiments

- As seen before, observed sequences are not, in essence, normally distributed. However, computer experiments show that the adjusted rescaled range r_N^{**} is seemingly insensitive to changes in the marginal distributions.
- This is the contrast with the adjusted range r_N^{*} which becomes independent of the distribution only when N is large, perhaps greater than 1000 in highly skewed data.
- The important characteristics of natural time series is persistence. This is the property by which high flows tend to follow high flows and low flows tend to follow low flows.
- A particular method of measuring persistence is through the serial correlagram. The effect of (positive) serial correlation on the adjusted rescaled range can be demonstrated by generating data sets using the AR(p) and the ARMA(p,q) models.
- In this way $K(N)$ and $h(N)$ are found to take on values greater than those for independent variates. These differences increase with increases in the serial correlation.
- However, they decrease for very large sample sizes; furthermore, it has been shown analytically that when N becomes very large the Hurst coefficient tends to the asymptotic value of 0.5 for ARMA processes.

Range analysis

Explanation of Hurst phenomenon and experiments

- It follows from all of this that the preasymptotic periods, over which the Hurst coefficient is greater than 0.5 are much larger than those from independent variates.
- From the practical point of view, because sample sizes investigated by Hurst were less than 2000, it seems reasonable to assume that serial correlation can, at least, partly account for the observed Hurst phenomenon.
- To conclude, it can be said that both serial correlation and non-stationarity can account for the Hurst effect over finite time horizons. Indeed, many natural processes seem to have both these properties in varying degrees so that it is difficult to identify the effect of each.